# Software Project Management

## M1 — Project Initiation Report

Project title: GoodTracker - Mobile Equipment Reservation Application

Team name/number: GoodTracker (course team number to confirm)

Team members and roles: Talha Yılmaz - mobile UI; Eylül Melda Çakmak - mobile integration and testing; Kerem Tuna - database and equipment API; Kardelen Dargın - reservation API and project tracking

Submission date: To be confirmed in the LMS

Document version: 0.1 - team review draft, 21 October 2026

Repository/project board link: See the Evidence appendix

## 1. Executive Summary

GoodTracker is our planned Android app for reserving shared laboratory and office equipment. Borrowers need a simple way to see what is available and request an item without creating overlapping bookings. Staff need a way to keep equipment records and decide on requests. For the first version, we will focus on sign-in, equipment browsing, availability, reservation requests and cancellation, and staff approval. Payments, university single sign-on, push notifications and an app-store release are outside this scope. In M1, we set the project objectives and boundaries, mapped the main stakeholder groups, chose a hybrid delivery approach, and listed success checks and initial risks. These are planning decisions; the app and its tests are not complete yet. The stakeholder needs are initial assumptions because no interview evidence is recorded. Our next steps are to confirm the course submission details, review this report as a team, and prepare the M2 backlog, estimates, schedule and quality plan. We will use repository commits and the project board to track actual work.

## 2. Project Charter

### 2.1 Objectives

| Objective | Success measure | Target |
|---|---|---|
| Find and reserve an item | Walk through sign-in, finding an item, availability, request and cancellation on Android. | The full flow works for a sample item. |
| Prevent double bookings | Submit two overlapping requests for one item and time slot. | At most one request is accepted. |
| Separate staff and user actions | Try staff actions with staff and regular user accounts. | Only staff manage items and approvals; users change only their own requests. |

*The targets above will be checked on a working build. They are not test results yet.*

## 2.2 Scope

| In scope | Out of scope |
|---|---|
| Sign-in, equipment browsing, search by category or location, and availability. | Payments and university single sign-on. |
| Reservation requests, cancellation, status, and conflict prevention. | Push notifications and an app-store release. |
| Staff equipment management and approval or rejection of requests. | A production rollout across campus. |

The proposed stack is React Native with Expo and TypeScript, Node.js with Express, and Supabase for data and sign-in. We will check it with a small prototype before committing to it.

## 2.3 Constraints

The work must fit the SE3003 milestones and a four-person student team. The first demo is limited to Android. The LMS deadline still needs checking. We do not yet have confirmed access to real equipment records or users, and no paid-service budget is recorded. The app and backend have not been built yet.

## 2.4 Assumptions

| ID | Assumption | If incorrect, what may happen? |
|---|---|---|
| A-01 | Students or staff need to see availability and request shared equipment; user feedback is still needed. | The target users or flow may change. |
| A-02 | A staff member will maintain item records and approve or reject requests. | Roles or approvals may need to change. |
| A-03 | Expo, Express and Supabase will support a small Android demo within available limits. | We may need to change the setup or scope. |
| A-04 | We can prepare sample equipment and reservation data. | Data preparation may take extra time. |

## 3. Stakeholder Analysis

| Stakeholder | Interest / Need | Influence | Engagement approach |
|---|---|---|---|
| Borrowers / end users | Availability and a clear request process (assumed) | High | Ask potential users to try a prototype and comment on the flow. |
| Inventory staff | Accurate item records and request decisions (assumed) | High | Check the proposed approval flow with a staff representative. |
| Development team | Clear scope, roles and decisions | High | Use issues, the board and reviews to coordinate work. |
| Instructor / TAs | Milestone evidence and course requirements | High | Submit the official report through the required course channel. |

Stakeholder map: Manage closely - borrowers, the development team and course staff. Keep satisfied - inventory staff. Keep informed - the wider campus community. Monitor - external service providers. Needs attributed to borrowers and inventory staff are assumptions until we collect feedback.

## 4. Delivery Approach Decision

We chose a hybrid approach. Fixed course milestones require planned reports and decisions, while our four-person team needs room to refine the mobile flow and API after feedback. A fully predictive plan would be hard to change; a purely agile plan would not cover the required baseline. We plan two-week work cycles and will set the detailed sprint plan in M2.

## 5. Initial Success Criteria

| Area | Criterion | How it will be verified |
|---|---|---|
| Product | No overlapping confirmed reservations for one item and time slot. | Submit overlapping requests and check that at most one succeeds. |
| Product | Regular users cannot use staff actions or change others' requests. | Try protected actions using a regular user account and check rejection. |
| Product | The Android demo supports the core reservation flow. | Walk through sign-in, browsing, availability and requesting on a named build. |
| Project management | The demo stays within the agreed Android, API and database scope. | Compare completed backlog items with the M1 scope. |
| Quality | Changes have a visible review and result record. | Check commits, reviews and test notes before closing work. |

*These are checks for later builds. No application test results are claimed in this M1 report.*

## 6. Initial Risk Register

| ID | Risk | Probability | Impact | Response | Owner | Status |
|---|---|---|---|---|---|---|
| R-01 | Exam and coursework load delays tasks | High | Medium | Keep tasks small and plan around exam weeks. | Kardelen | Open |
| R-02 | Supabase limits affect the demo | Medium | High | Try the stack early and track service usage. | Kerem | Open |
| R-03 | Two users book one slot | Medium | High | Use an atomic booking rule and test simultaneous requests. | Kardelen | Open |
| R-04 | Android screens display poorly | Medium | Medium | Check the demo on different screen sizes. | Eylül | Open |
| R-05 | Mobile work waits for API design | Medium | High | Agree API contracts and use sample responses early. | Talha | Open |
| R-06 | Extra features delay the core flow | Medium | Medium | Prioritize the MVP in M2 and defer extras. | Kerem | Open |

## 7. Planned vs. Actual

This comparison uses the assigned M1 outputs and repository history. We do not have a dated effort baseline, so it would be misleading to claim a schedule variance in hours or days.

| Planned activity | Actual result | Deviation | Reason | Corrective action |
|---|---|---|---|---|
| Project charter | Charter added to main (a3496ca). | Team review pending | The section was drafted first. | Review it in the shared report. |
| Stakeholder analysis | Talha added the stakeholder section (9301b96). | Needs unverified | No interview evidence is recorded. | Seek feedback on a prototype. |
| Success criteria | Eylül added criteria (4d9ba27). | Tests not run | No app build exists yet. | Run checks on a named build. |
| Delivery approach and risks | Kardelen added the section; PR #26 was merged. | Review record absent | No peer review is recorded on the PR. | Team to review this report. |
| M1 report | This Word and PDF draft combine the sections. | Review and submission pending | LMS details still need confirmation. | Confirm deadline, review, then submit. |

## 8. Next Milestone Plan

| Item | Owner | Target date | Expected evidence |
|---|---|---|---|
| Prioritized backlog and acceptance criteria | Talha | Before M2; LMS date to confirm | Backlog linked to issues (#6). |
| Effort estimates and assumptions | Kerem | Before M2; LMS date to confirm | Estimates and assumption log (#7). |
| Schedule, RACI and communication plan | Kardelen | Before M2; LMS date to confirm | Schedule and team documents (#8). |
| Quality plan and baseline report | Eylül | Before M2; LMS date to confirm | Quality plan and M2 report (#9). |

## 9. Individual Contributions

| Team member | Contribution | Evidence |
|---|---|---|
| Talha Yılmaz | Stakeholder analysis and map. | stakeholders.md; commit 9301b96 |
| Eylül Melda Çakmak | Initial success criteria and planned checks. | success-criteria.md; commit 4d9ba27 |
| Kardelen Dargın | Delivery approach and initial risk register. | approach-and-risks.md; commit f9e9e7e; PR #26 |
| Kerem Tuna | Project charter, risk wording alignment and coordination of this report draft. | charter.md; commits a3496ca and 6851aab; this report |

## Appendix: Evidence

The links below show the work available in the repository. Interview notes, executed tests and a team review record are not available yet.

Repository and project documents - current project source and M1 files.

Project board - task workflow and future progress tracking.

Charter commit - objectives, scope, constraints and assumptions.

Stakeholder commit - stakeholder analysis by Talha.

Success criteria commit - initial checks by Eylul.

Delivery approach PR - approach and risk section by Kardelen, merged into main.